MPG No. 2.2.4 $\bar{1}1'$ [Type II, triclinic] [G tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} G_x & G_y & G_z \end{bmatrix}$$

$$G_x = \mathbb{G}_x^{(1)}$$
$$G_y = \mathbb{G}_y^{(1)}$$
$$G_z = \mathbb{G}_z^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} G_{xxx} & G_{xxy} & G_{xxz} \\ G_{yyx} & G_{yyy} & G_{yyz} \\ G_{zzx} & G_{zzy} & G_{zzz} \\ G_{yzx} & G_{yyz} & G_{zzy} \\ G_{xxz} & G_{yzx} & G_{zzx} \\ G_{xxy} & G_{yyx} & G_{yzx} \end{bmatrix}$$

$$G_{xxx} = 3\mathbb{G}_x^{(1)} + 2\mathbb{G}_x^{\alpha(1)}$$
$$G_{xxy} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\beta(1)}$$
$$G_{xxz} = \mathbb{G}_z^{(1)} - \mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\beta(1)}$$
$$G_{yyx} = \mathbb{G}_x^{(1)} - \mathbb{G}_x^{\alpha(1)} + \mathbb{G}_x^{\beta(1)}$$
$$G_{yyy} = 3\mathbb{G}_y^{(1)} + 2\mathbb{G}_y^{\alpha(1)}$$
$$G_{yyz} = \mathbb{G}_z^{(1)} - \mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\beta(1)}$$
$$G_{zzx} = \mathbb{G}_x^{(1)} - \mathbb{G}_x^{\alpha(1)} - \mathbb{G}_x^{\beta(1)}$$
$$G_{zzy} = \mathbb{G}_y^{(1)} - \mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\beta(1)}$$
$$G_{zzz} = 3\mathbb{G}_z^{(1)} + 2\mathbb{G}_z^{\alpha(1)}$$
$$G_{yzx} = \mathbb{G}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & G_{xxy} & G_{xxz} \\ G_{yyx} & 0 & G_{yyz} \\ G_{zzx} & G_{zzy} & 0 \\ G_{yzx} & -\frac{G_{yyz}}{2} & -\frac{G_{zzy}}{2} \\ -\frac{G_{xxz}}{2} & G_{zxy} & -\frac{G_{zzx}}{2} \\ -\frac{G_{xxy}}{2} & -\frac{G_{yyx}}{2} & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$G_{xxy} = 2\mathbb{G}_{y}^{(2)} - 2\mathbb{Q}_{xz}^{(1)}$$
$$G_{xxz} = 2\mathbb{G}_{z}^{(2)} + 2\mathbb{Q}_{xy}^{(1)}$$
$$G_{yyx} = 2\mathbb{G}_{x}^{(2)} + 2\mathbb{Q}_{yz}^{(1)}$$
$$G_{yyz} = 2\mathbb{G}_{z}^{(2)} - 2\mathbb{Q}_{xy}^{(1)}$$
$$G_{zzx} = 2\mathbb{G}_{x}^{(2)} - 2\mathbb{Q}_{yz}^{(1)}$$
$$G_{zzy} = 2\mathbb{G}_{y}^{(2)} + 2\mathbb{Q}_{xz}^{(1)}$$
$$G_{yzx} = 3\mathbb{Q}_{u}^{(1)} + \mathbb{Q}_{v}^{(1)}$$
$$\frac{G_{yyz}}{2} = \mathbb{G}_{z}^{(2)} - \mathbb{Q}_{xy}^{(1)}$$
$$\frac{G_{zzy}}{2} = \mathbb{G}_{y}^{(2)} + \mathbb{Q}_{xz}^{(1)}$$
$$\frac{G_{xxz}}{2} = \mathbb{G}_{z}^{(2)} + \mathbb{Q}_{xy}^{(1)}$$
$$G_{zxy} = -3\mathbb{Q}_{u}^{(1)} + \mathbb{Q}_{v}^{(1)}$$
$$\frac{G_{zzx}}{2} = \mathbb{G}_{x}^{(2)} - \mathbb{Q}_{yz}^{(1)}$$
$$\frac{G_{xxy}}{2} = \mathbb{G}_{y}^{(2)} - \mathbb{Q}_{xz}^{(1)}$$
$$\frac{G_{yyx}}{2} = \mathbb{G}_{x}^{(2)} + \mathbb{Q}_{yz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & G_{yzy} & G_{yzz} \\ G_{zxx} & G_{zxy} & G_{zxz} \\ G_{xyx} & G_{xyy} & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = \mathbb{Q}_{0}^{(1)} - \mathbb{Q}_{u}^{(2)} + \mathbb{Q}_{v}^{(2)}$$
$$G_{yzy} = \mathbb{G}_{z}^{(3)} + \mathbb{Q}_{xy}^{(2)}$$
$$G_{yzz} = -\mathbb{G}_{y}^{(3)} + \mathbb{Q}_{xz}^{(2)}$$
$$G_{zxx} = -\mathbb{G}_{z}^{(3)} + \mathbb{Q}_{xy}^{(2)}$$
$$G_{zxy} = \mathbb{Q}_{0}^{(1)} - \mathbb{Q}_{u}^{(2)} - \mathbb{Q}_{v}^{(2)}$$
$$G_{zxz} = \mathbb{G}_{x}^{(3)} + \mathbb{Q}_{yz}^{(2)}$$
$$G_{xyx} = \mathbb{G}_{y}^{(3)} + \mathbb{Q}_{xz}^{(2)}$$
$$G_{xyy} = -\mathbb{G}_{x}^{(3)} + \mathbb{Q}_{yz}^{(2)}$$
$$G_{xyz} = \mathbb{Q}_{0}^{(1)} + 2\mathbb{Q}_{u}^{(2)}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).